



BIOLOGICAL SYSTEMS MODELING AND ANALYSIS

Homework Assignment 3

【Spring Semester, 2025】

Due Date: Apr. 16, 2026

Problem 1: (25%)

Modify and simulate the beer equations (Fig. 5.8) so that yeast growth uses the equation for Temperature Optimum shown in Fig. 5.4, curve A. Let temperature oscillate around 20°C with an amplitude of 8° and a period of 1 day with a peak at 12:00 noon. Discuss the effect of model parameters on the simulation results.

Please note that the primary fermentation takes about a week for the yeast to consume major part of sugar, and approximately 2.0665 g sugar yields 1 g ethanol, 0.9565 g CO₂, and 0.11 g yeast. The choice of model parameters should be compatible with these real conditions.

Some basic beer brewing information can be found from the following link:
<https://www.thekitchn.com/how-beer-is-brewed-a-timeline-the-kitchens-beer-school-2015-217246>.

Problem 2: (25%)

Non-dimensionalize the Lotka-Volterra equations (Eq. 4.23).

Problem 3: (25%)

Investigate the effect of RK-4 time steps on Eq. 6.4. Try $\Delta t = (1.0, 0.5, 0.1, 0.01)$. Continue to approximate the time step needed for the dynamics to converge.

Problem 4: (25%)

The following differential model can be set up to simulate the liquid level in a tank that has a constant flow rate in, but a cyclic flow rate out, where h represents the level and t the time.



$$16\sqrt{5h - h^2} \frac{dh}{dt} = -\sin \frac{\pi t}{60}; \quad h(0) = h_0$$

where $h_0 = 2.0$, $0.0 \leq t \leq 120$

Solve this model using both Euler and Runge-Kutta methods. How small must Δt be to prevent incorrect solution in the Euler method? How small in the RK method? Compare the results from the two methods.

Notes:

1. Please submit your homework report together with your program to the NTU COOL course website.
2. Late submission will have a penalty of 10% discount per day of your homework total score toward a maximum of 50% discount. No late submission over five days will be accepted.